

D.1 Gravitational fields

Topic warm-up · IB Physics (2023 syllabus)

with worked answers

Name: _____

Date: _____

Total: **27 marks**

Show your working.

Question 1 — The careless space elevator (8 marks)

A (fictional) space elevator carries tourists straight up from the surface of a planet of radius R , where the gravitational field strength is g_0 .

- (a) Define gravitational field strength at a point. [2]
- (b) Write down the field strength at the viewing deck, a height R above the surface. [2]
- (c) A tourist at that deck claims her weight is unchanged “because g is g ”. A second claims it is halved. State what fraction of her surface weight she actually experiences, and identify each tourist’s error. [3]
- (d) Write down the field strength at a distance $3R$ from the planet’s centre. [1]

Question 2 — The two moons of Zog (5 marks)

Moon Alpha orbits planet Zog in a circular orbit of radius r with a period of 8.0 days. Moon Beta orbits at radius $4r$.

- (a) State Kepler’s third law for circular orbits. [1]
- (b) Calculate the orbital period of Beta. [3]
- (c) An astrologer on Zog predicts 32 days, “four times the radius, four times the period”. State the exponent the astrologer used and the correct one. [1]

Question 3 — The double satellite (7 marks)

A communications satellite of mass m orbits a planet of mass M at radius r with speed v .

- (a) By equating gravitational force to the required centripetal force, show that $v = \sqrt{GM/r}$. [3]
- (b) A replacement satellite has mass $2m$ and is placed in the same orbit. State its orbital speed, and the feature of your equation in (a) that guarantees the answer. [2]
- (c) Hence explain why an astronaut and her spanner can float side by side in orbit, despite their very different masses. [2]

Question 4 — [HL] Which quantity am I?

(7 marks)

Four gravitational quantities introduce themselves at a party: *field strength*, *potential*, *potential energy* and *force*.

- (a) One guest says: “I am the work done per unit mass in bringing a small test mass from infinity to a point.” Name this quantity, and state its sign everywhere in the field. [3]
- (b) Give the defining phrase (in the same party style) for *field strength*, making the contrast explicit. [2]
- (c) An orbiting astronaut says she is in “zero gravity”. Using the idea of the field at her orbital radius, correct her in one or two sentences. [2]

Answers

Question 1 — The careless space elevator

- (a) The gravitational force per unit mass acting on a small test mass placed at that point. (Small, so it does not disturb the field it is measuring.)
- (b) A height R above the surface is a distance $r = 2R$ from the *centre*, which is what $1/r^2$ counts from. So $g = g_0 \left(\frac{R}{2R} \right)^2 = \frac{g_0}{4}$.
- (c) She experiences one *quarter* of her surface weight. The first tourist assumed the field is uniform, the ground-floor habit of treating g as a constant; that only holds for heights very small compared with R . The second scaled as $1/r$ instead of $1/r^2$, halving instead of quartering.
- (d) $g = g_0 \left(\frac{R}{3R} \right)^2 = \frac{g_0}{9}$.

Question 2 — The two moons of Zog

- (a) For bodies orbiting the same central mass, the square of the period is proportional to the cube of the orbital radius: $T^2 \propto r^3$.
- (b) $\frac{T_\beta}{T_\alpha} = \left(\frac{4r}{r} \right)^{3/2} = 4^{3/2} = 8$, so $T_\beta = 8.0 \times 8 = 64$ days.
- (c) The astrologer used an exponent of 1, taking $T \propto r$. The correct exponent is $\frac{3}{2}$, since $T \propto r^{3/2}$.

Question 3 — The double satellite

- (a) Equating the gravitational force to the centripetal force required:

$$\frac{GMm}{r^2} = \frac{mv^2}{r} \Rightarrow \frac{GM}{r} = v^2 \Rightarrow v = \sqrt{\frac{GM}{r}}$$

- (b) The *same* speed v . The satellite's own mass m cancelled in (a), so the expression contains only the planet's mass M and the radius r . Doubling m changes nothing.
- (c) Field strength is force per unit *mass*, so both the astronaut and the spanner are given the same acceleration by the planet's field regardless of their masses. They therefore follow the same orbit and stay alongside each other, both in continuous free fall.

Question 4 — [HL] Which quantity am I?

- (a) Gravitational *potential*, V_g . It is *negative* everywhere in the field, taking zero at infinity: the field does the work as a mass is brought inwards, so the work done per unit mass by an external agent is negative.
- (b) "I am the *force* per unit mass on a small test mass at the point." The contrast: field strength is force per kilogram, a vector, whereas potential is work (energy) per kilogram, a scalar.
- (c) The field at her orbital radius is not remotely zero; it is exactly what bends her straight-line motion into a circle. What she means is that she is unsupported, in continuous free fall with her spacecraft, so nothing pushes back on her. Weightless means unsupported, not field-free.